

FlowSync – Integration Failure Catalog

1. Overview

This document defines common failure patterns, OAuth issues, sync problems, and recovery steps for all external integrations supported by FlowSync.

It is used by support agents and AI systems to diagnose integration issues and guide users through resolution or reconnection workflows.

2. Supported Integrations

FlowSync currently supports the following third-party integrations:

- Slack
- Jira
- GitHub
- Google Drive
- Microsoft Teams

Each integration follows OAuth-based authentication and tokenized API access.

3. Slack Integration Issues

3.1 Common Failures

- Message delivery failure
- Channel sync delay
- Missing notifications

3.2 OAuth Issues

- Invalid authorization code
- Workspace mismatch
- Access denied by user

3.3 Sync Issues

- Delayed event updates
- Webhook failures
- Rate limit exceeded

3.4 Reconnect Steps

1. Go to FlowSync → Integrations
 2. Select Slack
 3. Click “Reconnect”
 4. Re-authorize workspace permissions
 5. Confirm connection status
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4. Jira Integration Issues

4.1 Common Failures

- Issue sync not working
- Ticket creation failure
- Project mapping mismatch

4.2 OAuth Issues

- Expired Jira token
- Incorrect site URL configuration
- Insufficient permissions for project access

4.3 Sync Issues

- Delayed issue updates
- Missing field mapping

- Webhook misconfiguration

4.4 Reconnect Steps

1. Navigate to Integrations → Jira
 2. Disconnect current instance
 3. Re-authenticate Atlassian account
 4. Select correct project scope
 5. Save configuration
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5. GitHub Integration Issues

5.1 Common Failures

- Repository sync failure
- Commit events not triggering
- Webhook delivery failure

5.2 OAuth Issues

- Invalid GitHub app permissions
- Repository access not granted
- Token revoked by user

5.3 Sync Issues

- Missing commits in activity feed
- Delay in pull request updates
- Branch filtering misconfiguration

5.4 Reconnect Steps

1. Open FlowSync integrations panel
2. Select GitHub
3. Reinstall GitHub App or reauthorize OAuth
4. Select repositories to sync

6. Google Drive Integration Issues

6.1 Common Failures

- File sync not working
- Permission denied errors
- Missing folder access

6.2 OAuth Issues

- Google consent not granted
- Token expired
- Drive scope not enabled

6.3 Sync Issues

- Delayed file updates
- Large file upload failures
- Shared drive access restrictions

6.4 Reconnect Steps

1. Go to Integrations → Google Drive
 2. Remove existing connection
 3. Re-authenticate Google account
 4. Grant required Drive permissions
 5. Re-sync folders
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7. Microsoft Teams Integration Issues

7.1 Common Failures

- Message notifications not delivered
- Channel mapping errors
- Bot not responding

7.2 OAuth Issues

- Azure AD consent not granted
- Tenant mismatch
- Expired access token

7.3 Sync Issues

- Delayed notifications
- Missing channel events
- Bot permission restrictions

7.4 Reconnect Steps

1. Navigate to Integrations → Microsoft Teams
 2. Reconnect Azure account
 3. Grant admin consent
 4. Reinstall FlowSync app in Teams
 5. Verify channel mapping
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8. Cross-Integration Failure Patterns

8.1 Token Expiry

- All integrations may fail if OAuth tokens expire
- System triggers automatic reconnect alerts

8.2 Permission Revocation

- User or admin revokes access from provider side
- Integration becomes inactive immediately

8.3 Rate Limiting

- API requests exceed provider limits
- Temporary sync delays occur

8.4 Webhook Failures

- External service fails to send event callbacks
 - Leads to missing real-time updates
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9. Standard Recovery Workflow

When any integration fails:

1. Detect failure via logs or user report
 2. Classify issue type (OAuth / Sync / Permissions)
 3. Attempt automatic retry if applicable
 4. Trigger reconnect instructions if needed
 5. Escalate to support or engineering if unresolved
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10. Escalation Conditions

Escalate to engineering when:

- Multiple integrations fail simultaneously
 - OAuth system-wide outage detected
 - Webhook infrastructure is down
 - Enterprise customer integrations fail repeatedly
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11. AI System Usage (Agentic Integration)

This document is used by AI agents to:

- Diagnose integration-specific failures
- Suggest correct reconnect workflows

- Identify OAuth and sync issues automatically
 - Reduce support ticket creation through self-healing steps
 - Escalate systemic integration outages
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12. Enhanced Integration Deep Dive (Detailed Behavior Models)

This section provides deeper behavioral and technical breakdowns for each integration to improve AI agent reasoning accuracy, failure diagnosis, and user guidance quality.

12.1 Slack – Deep Behavior Model

Core Purpose

Real-time team communication and automated notifications.

Key Event Types

- message_posted
- channel_created
- workflow_triggered
- alert_notification

Advanced Failure Patterns

- Event duplication due to retry loops
- Webhook signature mismatch
- Channel ID mapping drift after workspace update
- Bot message rate throttling during high load

Smart Diagnosis Rules

If Slack fails:

- If messages delayed → check webhook queue backlog
 - If messages missing → validate signing secret
 - If bot inactive → check app reinstall status
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12.2 Jira – Deep Behavior Model

Core Purpose

Project tracking, issue lifecycle management, and enterprise workflow coordination.

Key Event Types

- issue_created
- issue_updated
- status_changed
- sprint_synced

Advanced Failure Patterns

- Field mapping mismatch after schema update
- Issue duplication due to retry webhook
- Sprint sync desync in large projects
- Permission scope partial denial (project-level restriction)

Smart Diagnosis Rules

If Jira fails:

- If issue not created → validate API scope (write:issue)
 - If updates missing → check webhook event subscription
 - If project inaccessible → verify project-level permissions
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12.3 GitHub – Deep Behavior Model

Core Purpose

Source code collaboration, repository tracking, CI/CD event streaming.

Key Event Types

- push_event
- pull_request_opened
- pull_request_merged
- release_published

Advanced Failure Patterns

- Webhook replay causing duplicate commits
- Branch filtering misconfiguration
- GitHub App permission drift after org policy update
- Large repository sync delay due to payload size

Smart Diagnosis Rules

If GitHub fails:

- If commits missing → verify webhook delivery logs
 - If PR events missing → check repository selection scope
 - If delays occur → inspect rate limits and queue backlog
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12.4 Google Drive – Deep Behavior Model

Core Purpose

File storage, document collaboration, and workspace synchronization.

Key Event Types

- file_uploaded

- file_updated
- folder_shared
- permission_changed

Advanced Failure Patterns

- Drive API quota exhaustion
- Shared drive permission inheritance conflicts
- Large file chunk upload interruption
- Background sync collision across devices

Smart Diagnosis Rules

If Google Drive fails:

- If file missing → check folder permission scope
 - If upload fails → validate chunk size limits
 - If sync delay → inspect polling interval vs push event mode
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12.5 Microsoft Teams – Deep Behavior Model

Core Purpose

Enterprise communication, notifications, and workflow collaboration inside Microsoft ecosystem.

Key Event Types

- message_sent
- channel_event
- bot_interaction
- meeting_notification

Advanced Failure Patterns

- Tenant mismatch causing silent auth failure
- Bot permission degradation after admin policy update

- Delayed event propagation due to Graph API throttling
- Channel mapping inconsistency in multi-team environments

Smart Diagnosis Rules

If Teams fails:

- If bot not responding → verify Azure bot registration
 - If messages missing → check Graph API permissions
 - If tenant mismatch → validate enterprise directory ID
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13. Cross-Integration Intelligence Layer

Unified Failure Detection

The AI system should classify failures into:

- AUTH_LAYER_FAILURE (OAuth/token issues)
- SYNC_LAYER_FAILURE (event delay or missing updates)
- PERMISSION_LAYER_FAILURE (scope or access issue)
- INFRASTRUCTURE_FAILURE (rate limit or downtime)

Correlation Rules

- If multiple integrations fail simultaneously → suspect infrastructure or network issue
- If single integration fails → isolate provider-level issue
- If OAuth failures increase → token refresh system issue

Escalation Logic Enhancement

- 1 integration failing → standard support workflow
 - 2+ integrations failing → investigate shared dependency layer
 - All integrations failing → system-wide incident (P0)
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